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Prof. Dr. Matthias Schwab & Dr. Elke Schaeffeler
Dr. Margarete Fischer-Bosch Institute of Clinical Pharmacology
Robert-Bosch-Krankenhaus
Auerbachstraße 112
70376 Stuttgart

Post Doctoral Research Scientist - Bioinformatics (m/f/d)

Dear Prof. Dr. Schwab and Dr. Schaeffeler,

My experience refactoring production-grade medical data pipelines has taught me how crucial robust computational infrastructure is for clinical research. The opportunity to elucidate the tumor microenvironment by integrating spatial transcriptomics and scRNA-seq data at your institute directly matches this expertise. My transition from a Ph.D. in computational systems modeling to bioinformatics pipeline engineering makes me a highly suitable candidate to develop the novel computational solutions your clinical research demands.

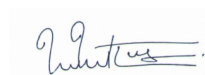
While my academic background is in systems modeling, I have spent the last year intensively retraining in modern bioinformatics, transcriptomics, and machine learning. At my current internship with HealthTwist GmbH, I am solely responsible for refactoring a production medical data pipeline (in R/tidyverse) for Germany's interventional radiology quality registry. Working with 143,000 patient records and 1,920 variables, I externalized hardcoded logic into configuration files, verified byte-level reproducibility, and reduced the 1,834-line codebase by 26%—establishing the robust biostatistics foundation required for integrating clinical phenotypes with multi-omics data.

Your project's need to bridge single-cell datasets with clinical outcomes requires deep expertise in both programming and data integration, which are my core strengths:

- **Bioinformatics & Transcriptomics Focus:** Highly proficient in building reproducible, containerized workflows (Nextflow/DSL2) and processing multi-omic data. I bring a strong understanding of the statistical modeling required to extract signal from high-dimensional biological datasets like spatial transcriptomics and scRNA-seq.
- **Strong Programming in R and Python:** I possess advanced programming skills across both ecosystems. In R, I build clinical data workflows (Tidymodels, Bioconductor, tidyverse) and perform complex biostatistics. In Python, I have engineered data analysis pipelines using Pandas, scikit-learn, and high-performance array frameworks (JAX), giving me the tools needed to prototype new computational solutions for data integration.
- **Large-Scale Data Integration & Oncology Interest:** Processed terabytes of raw experimental data during my Ph.D. and developed complex algorithms for HPC clusters. I am eager to apply my expertise in navigating coupled systems to cancer biology, specifically understanding metastasis and the tumor microenvironment in primary renal cell carcinoma.

I am deeply motivated by the Institute of Clinical Pharmacology's mission to bridge omics data with clinical phenotypes. My robust mathematical background and modern data engineering skills will allow me to immediately contribute to your interactive team of bioinformaticians, clinicians, and biologists. I welcome the opportunity to discuss further and am available for an interview.

Sincerely



Dr. Thejus Mahajan